

Quality metrics: what each number actually measures

Every fusion and registration algorithm in OpenFocus is tuned against the same set of numbers. This sheet says, in plain English, what each of those numbers means, which direction is better, and — the part that matters most — what each one *cannot* see. No metric here is a verdict on its own. They are used together precisely because each is blind to something another one catches.

The organising idea is **what a metric needs as input**. That decides where you are allowed to use it. A metric that needs a perfect answer to compare against can only run on synthetic test scenes; a metric that needs nothing but the result can run on any photograph you take.

5

METRIC FAMILIES

24

TERMS DEFINED

6 × 7

METHODS × SCENARIOS MEASURED

50

WORKED EXAMPLES

THE FIVE FAMILIES

- Full-reference** — needs the perfect answer. Only exists on synthetic scenes. *PSNR, SSIM, region PSNR, colour error.*
- No-reference** — needs the result and the source frames. Works on any real stack. *QABF, spatial frequency, entropy, std dev.*
- Artefact** — needs the result alone. Hunts one specific defect. *Seam excess/visible, defocus seams, block speckle, envelope.*
- Foreign-reference** — needs another program's render of the same capture. Not a truth, just a second opinion. *Detail agreement, detail share.*
- Geometric** — needs the true per-frame transform. Registration only, and it is exact. *Residual, gain, kept, selection share.*

WHAT EACH FAMILY CAN AND CANNOT BE RUN ON

	FULL-REFERENCE	NO-REFERENCE	ARTEFACT	FOREIGN	GEOMETRIC
Needs as input	result + the perfect image	result + the source frames	the result, nothing else	another render	true transforms
Where it can run	synthetic scenes only	every real stack	every real stack	real captures	handheld samples
Answers	"how close to correct?"	"did the detail survive?"	"is one defect present?"	"do we agree?"	"how many px off?"
Fails silently when	anything is shifted a pixel	noise is mistaken for detail	the defect is elsewhere	both are wrong	— it is exact

Read left to right and the trade-off is visible: the further right you go, the weaker the evidence, but the more real the picture it was measured on. Full-reference metrics are the most trustworthy numbers OpenFocus has and also the least representative — a rendered test scene is not a photograph of an insect. That is why nothing is decided on one family alone.

Where the numbers come from

the seven test scenarios

SCENARIOS

deep stack · depth edge · fine texture · long stack · low contrast · saturated colour · sensor noise. Each isolates one thing a fusion method can get wrong.

WHY SEVEN

A method that wins on texture and collapses on a depth edge has not won. Every observed range on this sheet spans all seven, which is why the spreads look wide.

The handheld sample scenes

where registration is measured

WHAT THEY HOLD

The exact transform applied to every frame, recorded when the scene was built. Alignment error is therefore computed, not judged.

CONSEQUENCE

Registration is the one part of the pipeline whose quality is known exactly. The numbers on page 6 are arithmetic; every other page is inference.

The regression baseline

how a change is judged

HOW IT WORKS

Every method's score on every scenario is recorded. A change that moves any of them by more than its tolerance has to be explained before it is accepted.

TOLERANCE

Listed per metric on page 7. It absorbs ordinary run-to-run variation — a move inside it is not a small win, it is no evidence at all.

FULL-REFERENCE METRICS

PSNR

Peak signal-to-noise ratio

higher better

dB

The average pixel error, rewritten as a number that grows as the error shrinks. Take how far every pixel is from where it should be, square it, average it, and put it on a logarithmic scale. Adding 6 dB means the error was cut in half.

GOES UP	The result matches the truth pixel for pixel. Infinite on an exact match.
GOES DOWN	Any deviation at all — blur, noise, a shifted edge, a colour cast.
BLIND TO	Where the error is. A faint softness spread over the whole frame and one ugly tear in the subject's eye can score identically. It has no idea what a picture is.
WATCH OUT	Needs pixel-perfect alignment. Shift the result one pixel and PSNR collapses, even though nothing looks worse. That is why it is useless against another program's render.

Observed across 6 fusion methods × 7 test scenarios

23.2 dB — worst median 35.9 69.4 dB — best

SSIM

Structural similarity index

higher better

-1 ... 1

Slides a small soft window over both images and, at every position, asks whether the local brightness, the local contrast and the local pattern agree. It asks “does this look like the same thing?” where PSNR asks “is this the same number?”.

GOES UP	Structure is preserved — edges land in the same place with the same contrast. 1.0 is identical.
GOES DOWN	Blur, ringing, halos, smearing — anything that changes local structure rather than overall level.
BLIND TO	Its own top end. Real results crowd into 0.97–1.00, so the difference between “good” and “excellent” is buried in the third decimal. Read the change, not the value.
USE IT FOR	Coherence. When a change makes the result hang together better without moving pixel error much, SSIM sees it and PSNR mostly does not.

Observed across 6 fusion methods × 7 test scenarios

0.894 — worst median 0.994 1.000 — best

Region PSNR

PSNR inside a boundary band

higher better

dB

The same PSNR, but computed only in a narrow strip — six pixels or so — hugging the borders where one depth meets another. That strip is where fusion has to make its hardest choice, and it is where halos and bleeding live.

WHY SEPARATE	The boundary is a tiny share of the frame. Averaged into a whole-frame score, a bad halo around every leg vanishes into a rounding error.
GOES DOWN	Bright fringes beside a dark subject, background bleeding into the subject, an edge taken from the wrong frame.
READ WITH	Whole-frame PSNR. Interior high and boundary low is the signature of a method that reconstructs flat areas well and mishandles depth edges.
NEEDS	The per-frame focus masks, so it is a synthetic-scene measurement.

Interpretation

boundary far below interior = halos equal = clean edges

Colour error

Mean absolute channel deviation

lower better

levels, 0–255

How far off the colour is, in the units your histogram is drawn in. Compare red with red, green with green, blue with blue, take the average distance. A value of 3 means the typical channel is three levels out of 255 away from correct.

WHY IT EXISTS	PSNR and SSIM are computed on brightness. Hues can drift a long way while both still look healthy — this is what “the colours came out wrong” measures as.
GOES UP	A fusion rule applied per channel that picks different frames for red and blue, or tone shifts from blending.
BLIND TO	Structure entirely. A perfectly graded but completely blurred result scores well here.
RULE OF THUMB	Under 1 level is invisible. Above 3–4 is a visible cast on smooth gradients.

Visibility

0 — exact ~3 — starts to show 10+ — obvious cast

Ground truth

the thing being compared against

The image the result *should* have been — every part of the subject sharp, and known to be correct because the scene was built rather than photographed.

WHERE FROM	Synthetic stacks: a sharp scene is rendered, then blurred by depth to manufacture the frames a camera would have taken. The unblurred original is the truth.
THE CATCH	A real focus stack has no such image. If it existed there would be nothing to fuse. That single fact is why the rest of this sheet exists.
SO	Everything on this page is a laboratory measurement — precise, trustworthy, and taken on a scene that is not a photograph of an insect.

Reading decibels

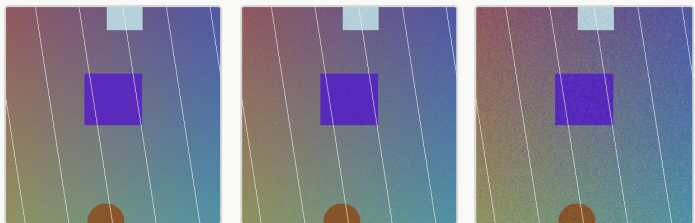
the dB scale, for PSNR

A dB is not a linear unit. Each step of about 6 dB means the error was *halved*. That is why 30 dB and 36 dB are not “close together”.

ROUGH GUIDE	Under 25 dB: visibly wrong. 30–35 dB: acceptable. Over 40 dB: differences are hard to see. Over 60 dB: essentially identical, and usually a flat, easy scene.
WHY 69 DB APPEARS	The low-contrast test scenario has almost nothing to get wrong. A spectacular PSNR there says the scene was easy, not that the method is good.
COMPARE	Only differences within one scenario. A 0.75 dB move is the point at which the regression suite treats a change as real.

PSNR: what the numbers look like

The same crop, degraded until it sits at a chosen PSNR. Each 6 dB step halves the error, so the gap between 34 and 26 dB is far larger than it sounds.



Ground truth PSNR ∞ · exact

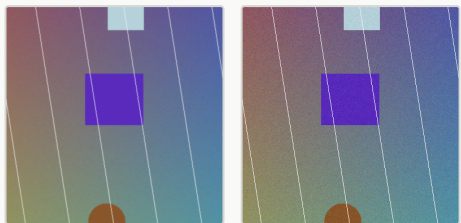
Slightly degraded PSNR 34.0 dB

Visibly degraded PSNR 26.0 dB

Under about 30 dB the damage is plain. Above 40 dB you will struggle to see it at all — which is why a difference of a few tenths of a dB decides nothing.

Why SSIM is quoted next to PSNR

Two degradations tuned to the same PSNR. If PSNR were the whole story these would be equally damaged. Look at them.



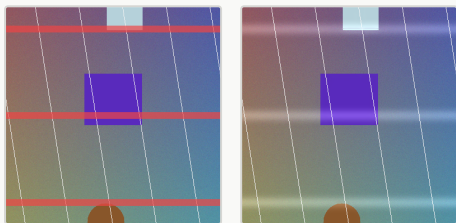
Blurred PSNR 30.1 · SSIM 0.929

Noised PSNR 30.0 · SSIM 0.758

PSNR calls them equal. SSIM separates them by 0.17 — a wide margin on a scale where real results live above 0.97. Structure is destroyed by noise in a way it is not by blur, and only one of the two metrics can tell.

Region PSNR: the boundary hides in the average

Left, the strip the metric scores — the band where two depths meet. Right, a render with a halo living in exactly that band.



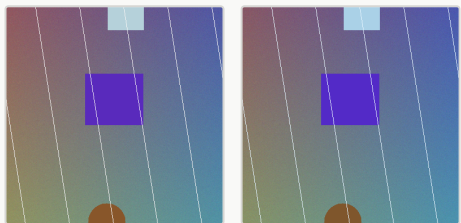
The boundary band 6 px either side of a depth edge

A haloed render whole frame 27.9 dB

The same render scores 27.9 dB over the whole frame and 17.0 dB inside the band. The defect is nearly eleven decibels worse than the frame average admits.

Colour error: what the brightness metrics miss

A cast applied to the channels, leaving every edge exactly where it was.



Correct colour reference

Warm cast error 5.0 levels

SSIM rates the cast version 0.998 — essentially perfect — because no structure moved. PSNR reads 31.9 dB. Only the colour metric reports the fault a viewer notices immediately.

NO-REFERENCE METRICS

QABF

Q^{AB/F} — gradient fusion quality
(Xydeas & Petrovic, 2000)

higher better

0 ... 1

Of all the edge detail that was present somewhere in the source frames, what fraction survived into the fused result? Strong edges count for more than faint ones, and an edge only counts if it kept both its *strength* and its *direction*.

WHY IT LEADS It needs no ground truth, so it is the headline number for real stacks — the one quoted when comparing methods on an actual capture.

GOES UP The method picked the sharp frame everywhere and did not soften the edge on its way through.

GOES DOWN Detail lost to blending, edges softened by smoothing, regions sourced from a defocused frame.

BLIND TO Colour, tone, and seams. It also cannot tell a *correct* edge from a *consistent* one: an edge faithfully copied from the wrong frame still scores.

Observed across 6 fusion methods × 7 test scenarios

0.336 — worst median 0.612 0.846 — best

Spatial frequency

RMS of the row and column gradients

higher usually better

levels

How busy the picture is. Measure how much brightness changes from each pixel to its neighbour going across, and going down, and combine the two. A blurred image barely changes between neighbours and scores low; a crisp one scores high.

GOES UP Retained fine detail and crisp edges.

ALSO GOES UP Sensor noise, block seams, speckle, ringing. Every defect that adds contrast between neighbouring pixels *raises* this score.

SO Never read it alone. Rising spatial frequency is good news only when the seam and speckle numbers on page 4 held still. A noisy render is the easiest way to top this metric.

SCALE Depends entirely on the subject. Compare methods on one scene, never one method across scenes.

Observed across 6 fusion methods × 7 test scenarios — note the 37× spread, it is subject-driven

2.4 — flat scene median 29.8 90.7 — fine texture

Entropy

Shannon entropy of the histogram

higher usually better

bits, 0 ... 8

How many different brightness values the picture actually uses, and how evenly. A flat grey card uses one value and carries almost no information: near 0 bits. A rich photograph spreads across the whole range: near 8 bits, the most an 8-bit image can hold.

GOES UP More tonal information preserved — detail that would otherwise be crushed into a single level survives.

ALSO GOES UP Noise, which is the purest form of “many different values”. Entropy cannot distinguish information from randomness.

BLIND TO Arrangement, completely. Shuffle every pixel in the frame and entropy does not move at all.

USE IT AS A supporting number: a drop in entropy alongside a drop in QABF confirms detail was genuinely lost rather than merely relocated.

Scale

0 — one flat tone ~7.5 typical photo 8 — theoretical max

Std dev

Grayscale standard deviation

higher usually better

levels

How far a typical pixel sits from the average brightness of the frame — a one-number stand-in for contrast. A washed-out result huddles around mid-grey and scores low; one with deep blacks and clean highlights scores high.

GOES DOWN Averaging across frames, which is what blending does when it cannot decide. This is the clearest symptom of a fusion rule that hedges.

BLIND TO Where the contrast came from. Clipping the blacks raises it. So does adding noise. So does a seam.

READ AS A sanity check, not a target. A large unexplained drop against the source frames means the method is washing the picture out. It costs nothing to compute, which is why it is logged everywhere — but it is the weakest evidence on this page.

Interpretation

below the source frames = washed out matching = contrast preserved

The noise trap

read this before quoting page 3

Three of the four metrics on this page — spatial frequency, entropy and std dev — go up when you add noise to the picture. All three would call a grainy render an improvement.

WHY Each is measuring “variation”. Detail is variation. So is noise. Nothing in the arithmetic separates them.

THE FAILURE Tune a method to maximise spatial frequency and you will end up with a sharpened, noisy result that scores beautifully and prints badly.

THE DEFENCE QABF is the exception: noise in the result matches no edge in any source frame, so it does not earn credit. Lead with QABF; use the other three to confirm, never to decide.

What “no reference” costs

the limit of this whole family

These metrics can tell you detail *survived*. None of them can tell you it ended up in the *right place*.

THE BLIND SPOT Take a sharp region from frame 12 and paste it where frame 30 should have won. The edges are all present and all crisp: QABF is content.

CAUGHT BY The artefact metrics on page 4, which know where a wrong choice has to leave its mark, and the geometric checks on page 6.

STILL, IT LEADS This is the only family that runs on the stacks people actually shoot. It is the headline evidence because it is the *available* evidence.

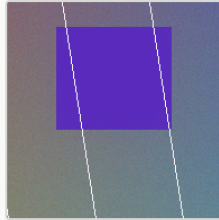
NO-REFERENCE METRICS, SEEN

QABF: how much of the edge detail arrived

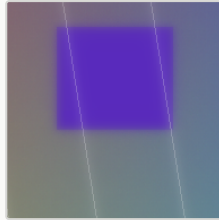
One frame of the stack, the stack fused properly, and the stack simply averaged together — the fusion rule that refuses to choose.



One source frame

QABF **0.299**

Fused (pyramid)

QABF **0.574**

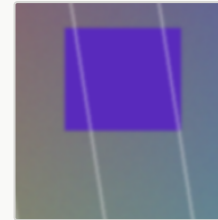
Averaged

QABF **0.173**

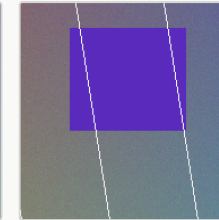
A single frame keeps only the edges that were sharp in it. Averaging keeps fewer than that. Fusing nearly doubles the single frame's score — and no ground truth was needed to say so.

The noise trap, in one row

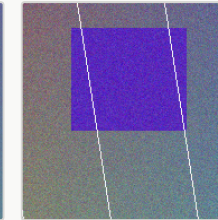
Soft, sharp, and sharp-plus-noise. **Read the middle number first**, then ask which picture you would want.



Softened

SF **3.1** · H 5.73

Sharp

SF **23.5** · H 5.52

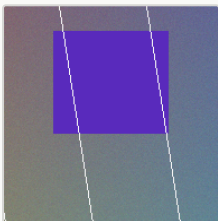
Sharp + noise

SF **27.6** · H 6.24

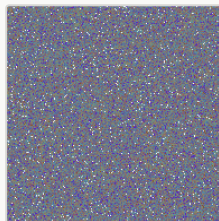
The noisy tile wins on **both** spatial frequency and entropy. Tune against either and you will be led straight to it. QABF is the one that refuses: it drops from **0.574** to **0.318**, because noise matches no edge in any source frame.

Entropy cannot see arrangement

The right-hand tile is the left-hand tile with **every pixel shuffled**. Not one pixel value was added or removed.



The picture

entropy **5.52** bits

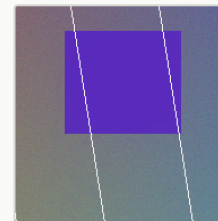
Same pixels, shuffled

entropy **5.52** bits

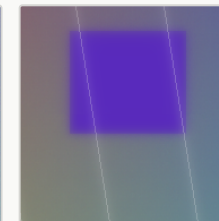
Identical entropy, to the second decimal, for a picture and for static. Spatial frequency at least notices — it doubles, from **23.5** to **47.7** — but it calls the static *better*.

Std dev: the cost of hedging

A method that chooses, against one that averages across the stack rather than commit to a frame.



Fused

std dev **30.0**

Averaged

std dev **25.6**

Averaging costs **15 %** of the contrast. That is the visible signature of a fusion rule that hedges — and the one thing this cheap metric is genuinely good for.

ARTEFACT METRICS — EACH HUNTS ONE SPECIFIC DEFECT

Seam excess

block_seams — part 1

lower better

How big a brightness step sits *on the block grid*, beyond the step you would expect from the picture there. In levels. Zero means the grid is invisible.

THE IDEA	Real detail does not know where a block-based method's lattice is. Only a fusion mistake lines up exactly with it.
APPLIES TO	Block-selection methods — DCT above all, where the block size is a tuning parameter this number responds to directly.
BLIND TO	Rarity. A hundred tiny steps and one enormous tear give the same mean — which is why the metric next to it exists.

Observed, all methods × scenarios

0.003 med 5.2 21.4

Seam visible

block_seams — part 2

lower better

%

What percentage of the block grid actually shows — the share of boundary pixels whose step is more than three times the local norm.

WHY PAIRED	Excess says <i>how bad</i> , visible says <i>how much</i> . One 100-level tear is a defect anyone notices; a hundred 1-level steps is not. A mean alone cannot tell them apart.
GOES UP	The lattice is showing across large parts of the frame.
READ AS	Low excess + low visible = clean. High excess + low visible = one bad tear. Low excess + high visible = a faint grid everywhere.

Regression tolerance: 2.0 percentage points

0 % — invisible 30 %+ — grid on show

Defocus seams

defocus_seams — excess & visible

lower better

The same two numbers, measured only in the parts of the picture that have no detail — the smooth, defocused background.

WHY RESTRICT	A seam is most objectionable exactly where there is nothing to hide behind. A method can score respectably overall while tearing the background to pieces.
SELF-LOCATING	The quiet region is found from the fused image itself, so this needs no reference and runs on any real stack.
SHARPEST TEST	Of the artefact set, this is the one that most reliably matches what a viewer complains about.

Observed excess, all methods × scenarios

0.0 med 0.32 25.7

Block speckle

block_speckle

lower better

%

The percentage of flat-area blocks standing far outside their neighbours — lone squares that came from the wrong frame.

THE DEFECT	A dust speck that happens to be sharp in exactly one frame wins its block. One square, wildly out of place, immediately obvious to a viewer.
WHY NEEDED	The seam metrics average along the grid — the wrong shape for an isolated square, which contributes almost nothing to a mean.
EARNED ITS PLACE	It was added after a mean-based seam score ranked a visibly speckled render the best of a parameter sweep.

Observed; tolerance 0.15 pp

0.0 % med 0.16 6.25 %

Envelope excursion

stack envelope check

must be zero

levels

How far the result strayed outside the darkest and brightest value any source frame had at that pixel. Anything above zero is a value the fusion invented.

PLAIN READING	If no frame ever recorded a pixel brighter than 180 there, a result of 205 did not come from the scene. It came from the algorithm.
SYMPTOM OF	Ringing and halos — the overshoot that transform-based methods produce beside a strong edge.
HARD GATE	In autotuning this is not scored, it is a veto. A candidate that leaves the envelope is rejected however well it scores on everything else.

Pass condition

(0, 0) — the only passing value

Why this family exists

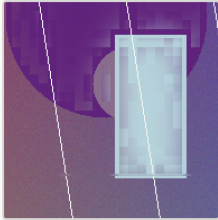
A general metric asks “is the picture good?” and answers by averaging over millions of pixels. A defect that occupies 0.2 % of the frame and ruins the print is, to an average, noise.

THE PATTERN	Each metric here encodes a <i>hypothesis</i> about where a specific algorithm puts its mistakes — on the block grid, in the flat background, outside the source range — and then looks only there.
CONSEQUENCE	They are not comparable between algorithm types. Seam metrics on a pyramid method measure nothing, because a pyramid has no lattice to tear along.
DIRECTION	All of them: lower is better, zero is the goal . Unlike everything on pages 2 and 3, there is no trade-off — nothing is lost by driving these to zero.

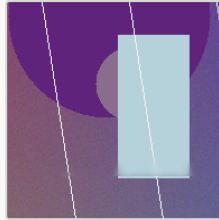
ARTEFACT METRICS, SEEN

Block seams, and the method that avoids them

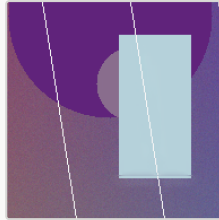
The same stack, three ways. **Naive** is textbook per-block winner-takes-all, which is the algorithm these metrics were written to catch — it is not what OpenFocus ships.



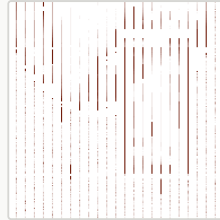
Naive block selection

visible **5.3 %** · 37.9 dB

DCT, block 8, kernel 7

visible **2.9 %** · 44.6 dB

Pyramid

visible **2.9 %** · 54.5 dB

Where the metric looks

steps on the lattice, ×14

The staircase along the naive edge is the defect. Both shipped methods sit at the **content floor of 2.9 %** — the score the pristine reference itself earns, since real detail sometimes lands on the lattice by chance.

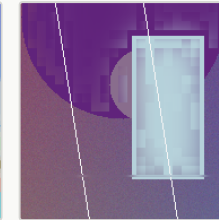
Defocus seams: the same defect, worse placed

Left, the region the metric restricts itself to — the quietest **40 %** of the frame, found on the render itself.

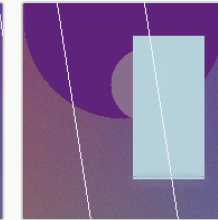


The quiet region

no detail to hide behind



Naive, in that region

visible **2.6 %**

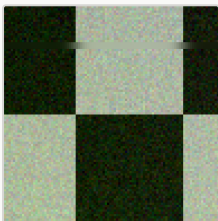
Pyramid, same window

visible **1.8 %**

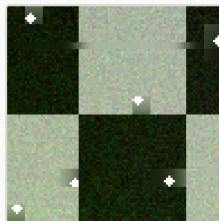
Needs no reference and no ground truth: the quiet region is measured on the fused image, so this one runs on any stack you shoot.

Speckle: one square from the wrong frame

Dust that is sharp in **exactly one frame** wins its block outright. Shown at 4×.



Clean stack

speckle **0.03 %**

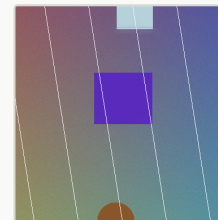
One dusty frame

speckle **0.76 %**

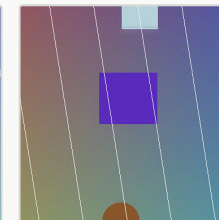
A twenty-five-fold rise, from a floor of **0.03 %**. To a mean-based seam score these squares are almost nothing; to a viewer they are the first thing visible.

Envelope: values that came from nowhere

A gentle unsharp mask — the amount nobody would think twice about — applied to a clean fusion. Right: how far each pixel escaped the source range.

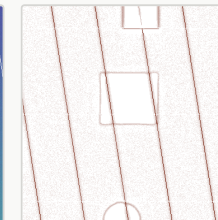


Fused, untouched

excursion **0** levels

After sharpening

+10 / -8 levels



Where it escaped

32.6 % of pixels

The clean fusion scores exactly **zero** — every pixel came from the stack. The sharpened one invents values at every edge. This is a veto, not a score, and it is easy to trip.

FOREIGN-REFERENCE METRICS

Tone matching

match_tone — a preparation step,
not a score

no direction

Before comparing anything, regrade one image onto the other's tone by matching their histograms, channel by channel. Another program's render of the same capture carries its own exposure, its own contrast curve, its own colour grade.

THE PROBLEM	None of that grading is a property of the fusion — but every intensity metric reads it as error, and a very large one. Uncorrected, the comparison measures taste, not quality.
WHAT IT REMOVES	Exactly the class of difference a global brightness curve can express, and nothing more. <i>Where</i> the detail sits cannot survive a per-channel lookup table, so real differences pass through untouched.
APPLIED	Per candidate, so a single source frame is regraded onto its own tone the same way the fused result is — neither is charged for a grade it never chose.
LIMIT	8-bit images only, which is the scale the metrics here are defined on.

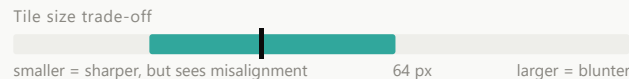
Detail map

detail_map — the shared building block

an image, not a number

A small picture of where the detail is: divide the frame into 64×64 pixel tiles and record the average local contrast in each. A photograph of a beetle becomes a coarse map with bright tiles on the resolved parts and dark tiles on the blur.

WHY TILES	Two programs' renders of one capture sit a few pixels apart — each aligned its own way, and focus breathing means no single warp relates them.
THE CONSEQUENCE	A pixel-by-pixel comparison would mostly measure that offset. A tile several times wider than the offset does not notice it.
THIS IS WHY	PSNR cannot be used against a foreign render at all, and the two metrics below can.



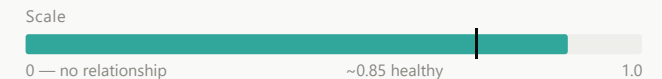
Detail agreement

detail_agreement — first return value

higher better -1 ... 1

How well the two detail maps correlate: do we and the other program agree about *which parts of the frame resolve*? At 1.0 both found detail in all the same places. At 0 the two disagree completely about what is sharp.

GOES DOWN	We resolved a region they left blurred, or — the worrying direction — they resolved something we did not.
BLIND TO	Amount. A heavily blurred copy of the reference keeps decent agreement while recovering no detail at all: the pattern of where detail <i>would</i> be is unchanged.
HENCE	Never quoted without the share, next card. Either one alone is trivially easy to satisfy.



Detail share

detail_agreement — second return value

1.0 is the target ratio

How much local contrast we recovered, as a fraction of theirs. 1.0 means as much detail as the other program pulled out of the same frames.

BELOW 1	Softer. Detail the capture contained is being lost somewhere in our pipeline.
ABOVE 1	Either genuinely better recovery or amplified noise and sharpening artefacts. An image of pure noise agrees with nothing yet scores an enormous share.
READ THE PAIR	High agreement <i>and</i> share near 1.0 is the only combination that means what it appears to mean.



Why PSNR is banned here

the two differences that are not errors

Two programs given the same capture return two images that differ in ways no metric should be counting — and that a pixel-comparison counts most heavily of all.

DIFFERENCE 1	Grade. Their exposure, contrast curve and colour rendering are choices, not quality. Removed by <i>tone matching</i> .
DIFFERENCE 2	Position. Each program aligned the stack its own way, and focus breathing means no single warp relates the two results. Removed by <i>pooling into tiles</i> .
LEFT OVER	Where the detail is, and how much of it there is. Those two survive both corrections — which is exactly why they are the two numbers reported.

The reference is a peer

how much weight to give it

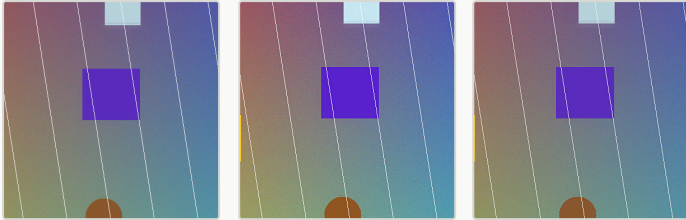
Another program's render is one more attempt at the same difficult problem. It is evidence, and it is the only evidence available on a real capture that comes from outside our own code — but it is not an answer key.

DISAGREEMENT MEANS	One of us handled that region better. Which one is a question to open, by looking at the region, not a verdict to record.
BOTH CAN BE WRONG	Nothing here detects a defect the two programs share. Two renders agreeing is not truth; it is a coincidence of approach.
ITS REAL VALUE	It catches the failures our own metrics are structurally blind to, because it was not designed around our assumptions.

FOREIGN-REFERENCE METRICS, SEEN

Tone matching: removing a difference that is not a fault

Their grade is darker and harder than ours. Nothing about that is a property of the fusion, but every intensity metric charges us for it.



Ours
the render being scored

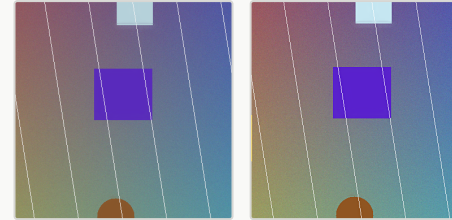
Theirs, as delivered
PSNR **28.2 dB**

Theirs, regraded
PSNR **43.6 dB**

Histogram matching alone recovers **15 dB**. That is how much of the apparent “error” was somebody else’s taste in contrast — and none of it was ever a fusion difference.

Why the comparison is pooled into tiles

Even after regrading, their render sits a few pixels from ours: each program aligned the stack its own way and no single warp relates the two.



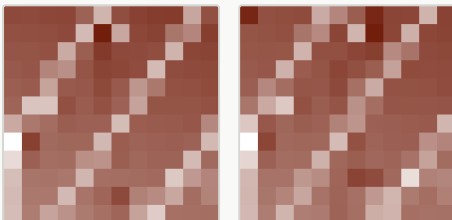
The render
regraded, still shifted

With the shift left in
PSNR **20.8 dB**

A five-pixel shift throws away the entire 15 dB the regrade recovered, and more. PSNR simply cannot be used here — which is the whole reason the detail map exists.

The detail map: where each render resolves

Local contrast pooled into **12 x 12** tiles of 64 px. Dark means detail. A few pixels of misalignment cannot move a tile this wide.



Our detail map
64 px tiles

Theirs, regraded
agreement **0.828**

The two maps agree at **0.828** with a share of **0.89**: we resolve the same places, at very nearly the same strength.

Why agreement is never quoted alone

A deliberately soft render, scored against the same foreign reference.



A soft render
agreement **0.672**

Its detail map
share **0.07**

Agreement barely falls — the soft render still knows *where* the detail belongs. The share collapses to **0.07**, because it recovered almost none of it. Either number alone would have missed this.

REGISTRATION METRICS — MEASURED IN PIXELS, NOT IMPRESSIONS

Residual — mean

mean px

lower better px

After alignment, how far a point of the scene still lands from where the reference frame put it — averaged over the whole stack. In output pixels.

PLAIN READING	1.2 px means a feature that should sit in one place is typically a pixel and a bit away from it in the other frames.
WHY EXACT	The sample scenes store the transform that was applied to each frame, so this is arithmetic, not an opinion about the picture.
MATTERS BECAUSE	Misalignment reaches fusion as double edges and ghosting — which the fusion metrics then blame on the fusion method.



Residual — worst

worst px

lower better px

The same measurement, for the single frame that ended up furthest out. The mean's companion.

WHY BOTH	A stack of forty frames, thirty-nine perfect and one badly misaligned, has an excellent mean. The output still has one ghosted layer running through it.
GOES UP	Feature matching failed on one frame — usually the ones furthest from the reference, where the least is in focus and there is least to match on.
READ AS	A wide gap between mean and worst points at a specific frame to look at, not at the pipeline as a whole.



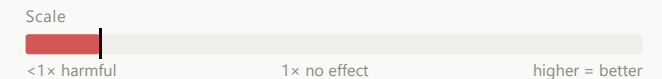
Gain

improvement factor

higher better x

The misalignment before registration divided by the misalignment after. 4× means the pipeline removed three quarters of the error.

WHY A RATIO	Raw pixel error depends on how badly the stack was shot. A ratio compares pipelines across scenes that were never equally shaky.
BELOW 1.0×	Registration made the alignment worse than leaving the frames alone. The benchmark flags this explicitly — it is a real outcome, not a hypothetical.
DIMINISHING	Each added stage costs time; gain is what says whether the stage paid for itself.



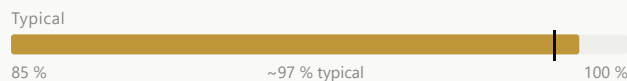
Kept

surviving frame area

higher better %

How much of the original frame is left after cropping to the region every aligned frame covers. The price of alignment, in field of view.

WHY IT DROPS	Shifting frames to line them up leaves each one missing a strip at some edge. Only the area they all cover can be kept.
THE TRADE	A more aggressive pipeline can align better and crop harder. 98 % kept at 0.6 px error may beat 91 % at 0.4 px, depending on how tightly the shot was framed.
NOT QUALITY	It measures cost, not correctness — but it is the number a photographer notices first.



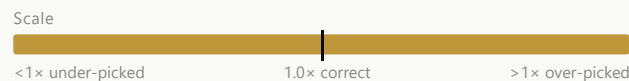
Selection share

vs. focus_index truth

closer to truth is better %

After registration, what share of the picture does the focus measure decide to take from the reference frame — and how does that compare with the share genuinely sharpest there?

WHAT IT TESTS	Whether alignment error is pushing the <i>focus decision</i> onto the wrong frame. That is a registration failure that looks exactly like a fusion failure.
REPORTED AS	A ratio against the ground-truth share. 1.0× means the measure is choosing frames as often as it should.
ABOVE 1	The reference frame is winning territory it does not deserve — a classic sign that the other frames are blurred by misalignment.



Detection width

a setting, not a score

tuning parameter px

The width the frames are scaled to *while the transform is being worked out*. The full-size frames are then warped by the result — they are never upsampled, so nothing is thrown away.

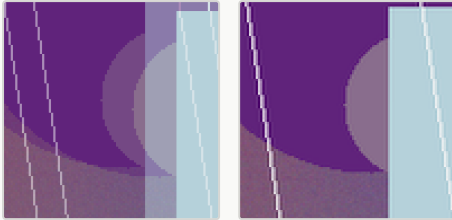
WHY IT EXISTS	Estimating a transform on a 60 MP frame is slow. Estimating it on a 2000 px version is fast and, up to a point, just as accurate.
TOO SMALL	Fine features vanish before matching sees them, and the residual rises.
ABOVE FRAME WIDTH	Means full-resolution detection: the setting is a cap, not a target.



REGISTRATION METRICS, SEEN

Residual: what misalignment looks like

Two frames of the stack laid over each other at half opacity. A misalignment has nowhere to hide in an overlay — it reads as a **double edge**. Shown at 4×.



Before registration

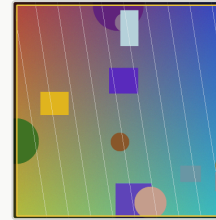
After registration

mean **12.81 px** · worst 14.56 mean **0.027 px** · worst 0.053

A gain of **469×**: from a visible double edge to under a twentieth of a pixel. Left uncorrected, every one of those double edges would have reached fusion — and the fusion metrics would have blamed the fusion.

Kept: what alignment costs in frame

Shifting frames to line them up leaves each one missing a strip at some edge. Only the area they all still cover can survive.



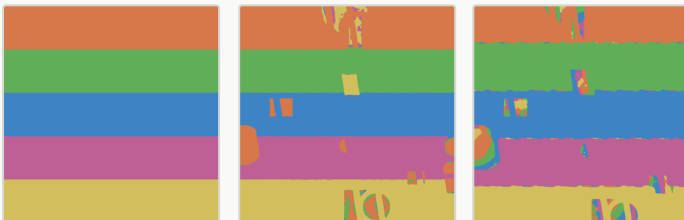
The surviving region

93.8 % · 742 x 746 px

The pipeline reported this crop itself. It is the price of the **469×** above, and the number a photographer notices before any other on this sheet.

Selection share: a registration fault wearing fusion's coat

Which frame the focus measure sources each pixel from. Colour = frame index. Left is the truth; the stack has five bands, so a correct answer is five clean stripes.



Ground truth

Aligned stack

Drift left in

which frame is genuinely sharpest

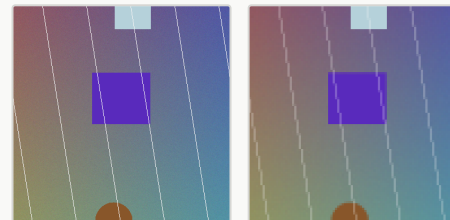
92.8 % correct

85.4 % correct

The drifted stack pulls whole regions from the wrong frame. Nothing about that render will look like a *registration* fault — it looks like the fusion picking badly.

Detection width: what the estimator is shown

The transform is worked out on a downscaled copy; the full-size frames are then warped by the result, never upsampled. Right shows what survives at a narrow detection width.



Full resolution

Detection at 192 px

every feature available

thin features gone

Nothing is lost from the *output* — only from what the matcher had to work with. Push the width too low and the fine features vanish before matching sees them, and the residual climbs.

REFERENCE CARD

TERM	FAMILY	BETTER	UNIT	OBSERVED RANGE	TOLERANCE	IN ONE LINE
PSNR Peak signal-to-noise ratio	 full-ref	higher	dB	23.2 – 69.4 (med 35.9)	0.75 dB	Average pixel error on a log scale. Blind to where the error is.
SSIM Structural similarity	 full-ref	higher	–1 ... 1	0.894 – 1.000 (med 0.994)	0.02	Does it look like the same thing locally. Saturates near 1.
Region PSNR boundary band	 full-ref	higher	dB	—	—	PSNR in a 6 px strip at depth edges. Catches halos the frame average hides.
Colour error mean abs. channel dev.	 full-ref	lower	levels 0–255	—	—	Hue drift in histogram units. What “the colours came out wrong” measures as.
QABF $Q^{AB/F}$ edge preservation	 no-ref	higher	0 ... 1	0.336 – 0.846 (med 0.612)	0.03	Share of source edges that survived. The headline metric for real stacks.
Spatial frequency row/col gradient RMS	 no-ref	higher	levels	2.4 – 90.7 (med 29.8)	1.0	How busy the picture is. Noise and seams raise it too.
Entropy Shannon, grayscale hist.	 no-ref	higher	bits 0 ... 8	~7.0 – 7.9 typical	—	How much tonal information is present. Blind to arrangement entirely.
Std dev grayscale spread	 no-ref	higher	levels	subject-dependent	—	Coarse contrast proxy. Drops when a method hedges and blends.
Seam excess block_seams · 1	 artefact	lower	levels	0.003 – 21.4 (med 5.2)	0.25	How big a step sits on the block grid, beyond the local norm.
Seam visible block_seams · 2	 artefact	lower	%	0 – 31 %	2.0 pp	How much of the grid actually shows. Excess says how bad; this says how much.
Defocus seam excess flat regions only	 artefact	lower	levels	0.0 – 25.7 (med 0.32)	0.25	Seams where there is no detail to hide behind — the most objectionable case.
Defocus seam visible flat regions only	 artefact	lower	%	0 – 31 % (med 2.7)	2.0 pp	Share of the grid showing in the smooth background.
Block speckle isolated wrong blocks	 artefact	lower	%	0.0 – 6.25 % (med 0.16)	0.15 pp	Lone squares taken from the wrong frame. Means cannot see them.
Envelope excursion stack min/max check	 artefact	must be 0	levels	(0, 0) required	veto	Values no source frame contained. Ringing and halos. A hard rejection.
Tone matching match_tone	 foreign	step	—	8-bit only	—	Strips out another program’s exposure and grade before comparing.
Detail agreement tile-map correlation	 foreign	higher	–1 ... 1	~0.85 healthy	—	Do we agree about which parts of the frame resolve. Survives misalignment.
Detail share contrast ratio	 foreign	1.0	ratio	0.7 – 1.3	—	How much local contrast we recovered vs. theirs. Above 1 may be noise.
Residual — mean geometric error	 geometric	lower	px	sub-pixel is the target	—	How far a scene point still lands from the reference, after alignment.
Residual — worst geometric error	 geometric	lower	px	—	—	The frame that got left behind. One bad frame ghosts the whole output.
Gain improvement factor	 geometric	higher	×	must exceed 1.0×	—	Error before ÷ error after. Below 1× registration made things worse.
Kept surviving area	 geometric	higher	%	~97 % typical	—	Field of view left after the common-region crop. Cost, not correctness.
Selection share vs. focus_index truth	 geometric	1.0×	%, ratio	—	—	Is misalignment pushing the focus choice onto the wrong frame.

Never read one number alone

Every metric on this sheet can be gamed by a defect. Spatial frequency and entropy both reward noise. SSIM rewards smoothing. Detail share rewards over-sharpening. A change is only an improvement when the metrics it *should not* have moved held still.

Compare within a scene, never across

Spatial frequency spans 2.4 to 90.7 across the test scenarios — a 37-fold range driven entirely by subject, not by method quality. Rank methods on one scene at a time. A number quoted without its scene means nothing.

Artefact scores have a floor

The pristine reference itself scores 2.4 on seam excess and 13 % on speckle, because real detail sometimes lands on the block lattice by chance. Only the *excess over that floor* is an artefact — an absolute artefact score, quoted without its floor, says nothing.

Tolerance is not precision

The tolerance column is how far a value may drift before the regression test treats it as a real change. It absorbs ordinary run-to-run variation. A move inside tolerance is not a small improvement — it is no evidence at all.